Name
Date
Course

Roots (Part III)

Perfect Cubes

1, 8, 27, 64, 125, 216, 343, 512, 729, 1000

Simplify.

① $\sqrt[3]{8} = \boxed{2}$

② $\sqrt[3]{64} = \boxed{4}$

③ $\sqrt[3]{27} = \boxed{3}$

④ $\sqrt[3]{125} = \boxed{5}$

⑤ $\sqrt[3]{-216} = \boxed{-6}$

⑥ $\sqrt[3]{-343} = \boxed{-7}$

n^{th} Root Rules

$\sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$

$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$

⑦ $\sqrt[3]{16} = \sqrt[3]{8\cdot 2} = \sqrt[3]{8}\cdot\sqrt[3]{2} = \boxed{2\sqrt[3]{2}}$

⑧ $\sqrt[3]{3000} = \sqrt[3]{1000\cdot 3} = \sqrt[3]{1000}\cdot\sqrt[3]{3} = \boxed{10\sqrt[3]{3}}$

⑨ $\sqrt[3]{-80} = \sqrt[3]{-8\cdot 10} = \sqrt[3]{-8}\cdot\sqrt[3]{10} = \boxed{-2\sqrt[3]{10}}$

⑩ $\sqrt[3]{24} = \sqrt[3]{8\cdot 3} = \sqrt[3]{8}\cdot\sqrt[3]{3} = \boxed{2\sqrt[3]{3}}$

⑪ $\sqrt[3]{-81} = \sqrt[3]{-27\cdot 3} = \sqrt[3]{-27}\cdot\sqrt[3]{3} = \boxed{-3\sqrt[3]{3}}$

⑫ $\sqrt[3]{128} = \sqrt[3]{64\cdot 2} = \sqrt[3]{64}\,\sqrt[3]{2} = \boxed{4\sqrt[3]{2}}$

⑬ $\sqrt[3]{x^3} = \boxed{x}$

⑭ $\sqrt[3]{x^9} = \boxed{x^3}$

⑮ $\sqrt[3]{x^6} = \boxed{x^2}$

⑯ $\sqrt[3]{x^{12}} = \boxed{x^4}$

⑰ $\sqrt[3]{x^5} = \sqrt[3]{x^3\cdot x^2} = \sqrt[3]{x^3}\cdot\sqrt[3]{x^2} = \boxed{x\sqrt[3]{x^2}}$

⑱ $\sqrt[3]{x^{10}} = \sqrt[3]{x^9\cdot x} = \sqrt[3]{x^9}\cdot\sqrt[3]{x} = \boxed{x^3\sqrt[3]{x}}$

⑲ $\sqrt[3]{x^4} = \sqrt[3]{x^3\cdot x} = \sqrt[3]{x^3}\cdot\sqrt[3]{x} = \boxed{x\sqrt[3]{x}}$

⑳ $\sqrt[3]{x^8} = \sqrt[3]{x^6\cdot x^2} = \sqrt[3]{x^6}\cdot\sqrt[3]{x^2} = \boxed{x^2\sqrt[3]{x^2}}$

㉑ $\sqrt[3]{250x^{13}} = \sqrt[3]{125\cdot 2\cdot x^{12}\cdot x} = \sqrt[3]{125}\cdot\sqrt[3]{2}\cdot\sqrt[3]{x^{12}}\cdot\sqrt[3]{x}$

$= 5\sqrt[3]{2}\,x^4\sqrt[3]{x}$

$= \boxed{5x^4\sqrt[3]{2x}}$

㉒ $\sqrt[3]{-40x^6} = \sqrt[3]{-8\cdot 5\cdot x^6} = \sqrt[3]{-8}\,\sqrt[3]{5}\,\sqrt[3]{x^6} = -2\sqrt[3]{5}\,x^2$

$= \boxed{-2x^2\sqrt[3]{5}}$

㉓ $\sqrt[3]{72x^5} = \sqrt[3]{8\cdot 9\cdot x^3\cdot x^2} = \sqrt[3]{8}\cdot\sqrt[3]{9}\cdot\sqrt[3]{x^3}\cdot\sqrt[3]{x^2}$

$= 2\cdot\sqrt[3]{9}\cdot x\sqrt[3]{x^2}$

$= \boxed{2x\sqrt[3]{9x^2}}$

㉔ $\sqrt[3]{-192x^7} = \sqrt[3]{-64\cdot 3\cdot x^6\cdot x} = \sqrt[3]{-64}\cdot\sqrt[3]{3}\cdot\sqrt[3]{x^6}\cdot\sqrt[3]{x}$

$= -4\sqrt[3]{3}\cdot x^2\cdot\sqrt[3]{x}$

$= \boxed{-4x^2\sqrt[3]{3x}}$

(25) $\sqrt[3]{16x^2}\cdot\sqrt[3]{2x^7} = \sqrt[3]{16x^2\cdot 2x^7} = \sqrt[3]{32x^9} = \sqrt[3]{8\cdot 4\cdot x^9}$

$= \sqrt[3]{8}\cdot\sqrt[3]{4}\cdot\sqrt[3]{x^9}$

$= 2\sqrt[3]{4}\cdot x^3$

$= \boxed{2x^3\sqrt[3]{4}}$

(26) $6\sqrt[3]{25x^4}\cdot\sqrt[3]{10x^2} = 6\sqrt[3]{25x^4\cdot 10x^2} = 6\sqrt[3]{250x^6}$

$= 6\sqrt[3]{125\cdot 2\cdot x^6}$

$= 6\sqrt[3]{125}\sqrt[3]{2}\sqrt[3]{x^6}$

$= 6\cdot 5\cdot\sqrt[3]{2}\cdot x^2$

$= \boxed{30x^2\sqrt[3]{2}}$

(27) $\sqrt[3]{12x^2}\cdot\sqrt[3]{9x^{10}} = \sqrt[3]{12x^2\cdot 9x^{10}} = \sqrt[3]{108x^{12}}$

$= \sqrt[3]{27\cdot 4\cdot x^{12}}$

$= \sqrt[3]{27}\cdot\sqrt[3]{4}\cdot\sqrt[3]{x^{12}}$

$= 3\sqrt[3]{4}\cdot x^4$

$= \boxed{3x^4\sqrt[3]{4}}$

(28) $\sqrt[3]{20x^2}\cdot\sqrt[3]{6x^5} = \sqrt[3]{20x^2\cdot 6x^5} = \sqrt[3]{120x^7} = \sqrt[3]{8\cdot 15\cdot x^6\cdot x}$

$= \sqrt[3]{8}\sqrt[3]{15}\sqrt[3]{x^6}\sqrt[3]{x}$

$= 2\sqrt[3]{15}\,x^2\sqrt[3]{x}$

$= \boxed{2x^2\sqrt[3]{15x}}$

(29) $2\sqrt[3]{48x^5} - 6x\sqrt[3]{-162x^2}$

$2\sqrt[3]{8\cdot 6\cdot x^3\cdot x^2} - 6x\sqrt[3]{-27\cdot 6\cdot x^2}$

$2\sqrt[3]{8}\sqrt[3]{6}\sqrt[3]{x^3}\sqrt[3]{x^2} - 6x\sqrt[3]{-27}\sqrt[3]{6}\sqrt[3]{x^2}$

$2\cdot 2\cdot\sqrt[3]{6}\cdot x\cdot\sqrt[3]{x^2} - 6x(-3)\sqrt[3]{6}\sqrt[3]{x^2}$

$4x\sqrt[3]{6x^2} + 18x\sqrt[3]{6x^2}$

$\boxed{22x\sqrt[3]{6x^2}}$

(30) $-9x\sqrt[3]{4} + 4\sqrt[3]{32x^3}$

$-9x\sqrt[3]{4} + 4\sqrt[3]{8\cdot 4\cdot x^3}$

$-9x\sqrt[3]{4} + 4\sqrt[3]{8}\sqrt[3]{4}\sqrt[3]{x^3}$

$-9x\sqrt[3]{4} + 4\cdot 2\cdot\sqrt[3]{4}\cdot x$

$-9x\sqrt[3]{4} + 8x\sqrt[3]{4}$

$\boxed{-x\sqrt[3]{4}}$

(31) $\dfrac{-4}{\sqrt[3]{9}}$

$\dfrac{-4}{\sqrt[3]{9}}\cdot\dfrac{\sqrt[3]{3}}{\sqrt[3]{3}}$

$\dfrac{-4\sqrt[3]{3}}{\sqrt[3]{9\cdot 3}}$

$\dfrac{-4\sqrt[3]{3}}{\sqrt[3]{27}}$

$\boxed{\dfrac{-4\sqrt[3]{3}}{3}}$

(32) $\dfrac{8}{\sqrt[3]{5}}$

$\dfrac{8}{\sqrt[3]{5}}\cdot\dfrac{\sqrt[3]{25}}{\sqrt[3]{25}}$

$\dfrac{8\sqrt[3]{25}}{\sqrt[3]{5\cdot 25}}$

$\dfrac{8\sqrt[3]{25}}{\sqrt[3]{125}}$

$\boxed{\dfrac{8\sqrt[3]{25}}{5}}$

(33) $\dfrac{5}{\sqrt[3]{2}}$

$\dfrac{5}{\sqrt[3]{2}}\cdot\dfrac{\sqrt[3]{4}}{\sqrt[3]{4}}$

$\dfrac{5\sqrt[3]{4}}{\sqrt[3]{2\cdot 4}}$

$\dfrac{5\sqrt[3]{4}}{\sqrt[3]{8}}$

$\boxed{\dfrac{5\sqrt[3]{4}}{2}}$

(34) $\dfrac{1}{\sqrt[3]{25}}$

$\dfrac{1}{\sqrt[3]{25}}\cdot\dfrac{\sqrt[3]{5}}{\sqrt[3]{5}}$

$\dfrac{\sqrt[3]{5}}{\sqrt[3]{125}}$

$\boxed{\dfrac{\sqrt[3]{5}}{5}}$

(35) $\dfrac{2}{\sqrt[3]{24}}$

$\dfrac{2}{\sqrt[3]{8\cdot 3}}$

$\dfrac{2}{\sqrt[3]{8}\cdot\sqrt[3]{3}}$

$\dfrac{2}{2\sqrt[3]{3}}$

$\dfrac{1}{\sqrt[3]{3}}$

$\dfrac{1}{\sqrt[3]{3}}\cdot\dfrac{\sqrt[3]{9}}{\sqrt[3]{9}}$

$\dfrac{\sqrt[3]{9}}{\sqrt[3]{3\cdot 9}}$

$\dfrac{\sqrt[3]{9}}{\sqrt[3]{27}}$

$\boxed{\dfrac{\sqrt[3]{9}}{3}}$

(36) $\dfrac{9}{\sqrt[3]{7000}}$

$\dfrac{9}{\sqrt[3]{1000\cdot 7}}$

$\dfrac{9}{\sqrt[3]{1000}\cdot\sqrt[3]{7}}$

$\dfrac{9}{10\sqrt[3]{7}}$

$\dfrac{9}{10\sqrt[3]{7}}\cdot\dfrac{\sqrt[3]{49}}{\sqrt[3]{49}}$

$\dfrac{9\sqrt[3]{49}}{10\sqrt[3]{343}}$

$\dfrac{9\sqrt[3]{49}}{10\cdot 7}$

$\boxed{\dfrac{9\sqrt[3]{49}}{70}}$

(37) $\sqrt[3]{\dfrac{5}{27}}$

$\dfrac{\sqrt[3]{5}}{\sqrt[3]{27}}$

$\boxed{\dfrac{\sqrt[3]{5}}{3}}$

(38) $\sqrt[3]{\dfrac{11}{512}}$

$\dfrac{\sqrt[3]{11}}{\sqrt[3]{512}}$

$\boxed{\dfrac{\sqrt[3]{11}}{8}}$

(39) $\sqrt[3]{\dfrac{729}{7}}$

$\dfrac{\sqrt[3]{729}}{\sqrt[3]{7}}$

$\dfrac{9}{\sqrt[3]{7}}$

$\dfrac{9}{\sqrt[3]{7}}\cdot\dfrac{\sqrt[3]{49}}{\sqrt[3]{49}}$

$\dfrac{9\sqrt[3]{49}}{\sqrt[3]{7\cdot 49}}$

$\dfrac{9\sqrt[3]{49}}{\sqrt[3]{343}}$

$\boxed{\dfrac{9\sqrt[3]{49}}{7}}$

(40) $\sqrt[3]{\dfrac{512}{5}}$

$\dfrac{\sqrt[3]{512}}{\sqrt[3]{5}}$

$\dfrac{8}{\sqrt[3]{5}}$

$\dfrac{8}{\sqrt[3]{5}}\cdot\dfrac{\sqrt[3]{25}}{\sqrt[3]{25}}$

$\dfrac{8\sqrt[3]{25}}{\sqrt[3]{5\cdot 25}}$

$\dfrac{8\sqrt[3]{25}}{\sqrt[3]{125}}$

$\boxed{\dfrac{8\sqrt[3]{25}}{5}}$

Fourth Powers

$1^4, 2^4, 3^4, 4^4, \ldots$

$1, 16, 81, 256, \ldots$

(41) $\sqrt[4]{16} = \boxed{2}$

(42) $\sqrt[4]{81} = \boxed{3}$

* Assume that all variables are positive.

(43) $\sqrt[4]{x^4} = \boxed{x}$

(44) $\sqrt[4]{x^{12}} = \boxed{x^3}$

(45) $\sqrt[4]{x^8} = \boxed{x^2}$

(46) $\sqrt[4]{x^{16}} = \boxed{x^4}$

(47) $\sqrt[4]{x^{20}} = \boxed{x^5}$

(48) $\sqrt[4]{x^{10}} = \sqrt[4]{x^8 \cdot x^2}$
$= \sqrt[4]{x^8} \cdot \sqrt[4]{x^2}$
$= \boxed{x^2 \sqrt[4]{x^2}}$

(49) $\sqrt[4]{x^5} = \sqrt[4]{x^4 \cdot x} = \sqrt[4]{x^4} \cdot \sqrt[4]{x} = \boxed{x\sqrt[4]{x}}$

(50) $\sqrt[4]{\dfrac{3}{8}}$

$$\frac{\sqrt[4]{3}}{\sqrt[4]{8}}$$

$$\frac{\sqrt[4]{3} \cdot \sqrt[4]{2}}{\sqrt[4]{8} \cdot \sqrt[4]{2}}$$

$$\frac{\sqrt[4]{3 \cdot 2}}{\sqrt[4]{8 \cdot 2}}$$

$$\frac{\sqrt[4]{6}}{\sqrt[4]{16}} = \boxed{\frac{\sqrt[4]{6}}{2}}$$

(51) $\sqrt[4]{\dfrac{5}{9}}$

$$\frac{\sqrt[4]{5}}{\sqrt[4]{9}}$$

$$\frac{\sqrt[4]{5} \cdot \sqrt[4]{9}}{\sqrt[4]{9} \cdot \sqrt[4]{9}}$$

$$\frac{\sqrt[4]{5 \cdot 9}}{\sqrt[4]{9 \cdot 9}}$$

$$\frac{\sqrt[4]{45}}{\sqrt[4]{81}} = \boxed{\frac{\sqrt[4]{45}}{3}}$$

(52) $\sqrt[4]{\dfrac{7}{48}}$

$$\frac{\sqrt[4]{7}}{\sqrt[4]{48}}$$

$$\frac{\sqrt[4]{7}}{\sqrt[4]{16 \cdot 3}}$$

$$\frac{\sqrt[4]{7}}{\sqrt[4]{16} \cdot \sqrt[4]{3}}$$

$$\frac{\sqrt[4]{7}}{2\sqrt[4]{3}}$$

$$\frac{\sqrt[4]{7} \cdot \sqrt[4]{27}}{2\sqrt[4]{3} \cdot \sqrt[4]{27}}$$

$$\frac{\sqrt[4]{7 \cdot 27}}{2\sqrt[4]{3 \cdot 27}}$$

$$\frac{\sqrt[4]{189}}{2\sqrt[4]{81}}$$

$$\frac{\sqrt[4]{189}}{2 \cdot 3} = \boxed{\frac{\sqrt[4]{189}}{6}}$$

(53) $\sqrt[4]{\dfrac{64x^6}{9}}$

$$\frac{\sqrt[4]{64x^6}}{\sqrt[4]{9}}$$

$$\frac{\sqrt[4]{16 \cdot 4 \cdot x^4 \cdot x^2}}{\sqrt[4]{9}}$$

$$\frac{\sqrt[4]{16} \cdot \sqrt[4]{4}\,\sqrt[4]{x^4}\,\sqrt[4]{x^2}}{\sqrt[4]{9}}$$

$$\frac{2 \cdot \sqrt[4]{4} \cdot x\sqrt[4]{x^2}}{\sqrt[4]{9}}$$

$$\frac{2x\sqrt[4]{4x^2}}{\sqrt[4]{9}}$$

$$\frac{2x\sqrt[4]{4x^2} \cdot \sqrt[4]{9}}{\sqrt[4]{9} \cdot \sqrt[4]{9}}$$

$$\frac{2x\sqrt[4]{4x^2 \cdot 9}}{\sqrt[4]{9 \cdot 9}}$$

$$\frac{2x\sqrt[4]{36x^2}}{\sqrt[4]{81}} = \boxed{\frac{2x\sqrt[4]{36x^2}}{3}}$$

(54) $$\sqrt[4]{\frac{32}{9}}$$

$$\frac{\sqrt[4]{32}}{\sqrt[4]{9}}$$

$$\frac{\sqrt[4]{16\cdot 2}}{\sqrt[4]{9}}$$

$$\frac{\sqrt[4]{16}\sqrt[4]{2}}{\sqrt[4]{9}}$$

$$\frac{2\sqrt[4]{2}}{\sqrt[4]{9}}$$

$$\frac{2\sqrt[4]{2}\ \sqrt[4]{9}}{\sqrt[4]{9}\cdot\sqrt[4]{9}}$$

$$\frac{2\sqrt[4]{2\cdot 9}}{\sqrt[4]{9\cdot 9}}$$

$$\frac{2\sqrt[4]{18}}{\sqrt[4]{81}}$$

$$\boxed{\frac{2\sqrt[4]{18}}{3}}$$

(55) $$\sqrt[4]{\frac{2}{243x^9}}$$

$$\frac{\sqrt[4]{2}}{\sqrt[4]{243x^9}}$$

$$\frac{\sqrt[4]{2}}{\sqrt[4]{81\cdot 3\cdot x^8\cdot x}}$$

$$\frac{\sqrt[4]{2}}{\sqrt[4]{81}\sqrt[4]{3}\sqrt[4]{x^8}\sqrt[4]{x}}$$

$$\frac{\sqrt[4]{2}}{3\sqrt[4]{3}\cdot x^2\sqrt[4]{x}}$$

$$\frac{\sqrt[4]{2}}{3x^2\sqrt[4]{3x}}$$

$$\frac{\sqrt[4]{2}\ \sqrt[4]{27x^3}}{3x^2\sqrt[4]{3x}\cdot\sqrt[4]{27x^3}}$$

$$\frac{\sqrt[4]{2\cdot 27x^3}}{3x^2\sqrt[4]{3x\cdot 27x^3}}$$

$$\frac{\sqrt[4]{54x^3}}{3x^2\sqrt[4]{81x^4}} = \frac{\sqrt[4]{54x^3}}{3x^2\sqrt[4]{81}\ \sqrt[4]{x^4}}$$

$$= \frac{\sqrt[4]{54x^3}}{3x^2\cdot 3\cdot x} = \boxed{\frac{\sqrt[4]{54x^3}}{9x^3}}$$